

UPMANYU JHA

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EXPERIENCE

Machine Learning Engineer (LLMs, NLP, CV) | *Quantiphi Analytics Solutions Private Limited* | Mumbai **Jul 2023 - Present**

- Led the end-to-end delivery of **15+ generative modeling pipelines** and **5 major Full-Stack** solutions on Google Cloud (Vertex-AI, Document-AI)
- Processed **80k+ documents** quarterly and boosted the F1-score for classification tasks by **19.5%** through model and pipeline optimization
- Eliminated model **hallucinations** by **85%**, cut **GCP Cost** by **15%** and gained a **92%** average F1-score for critical fields from structured data from diverse unstructured documents for reducing manual processing time by over **40%**
- Implemented **generative multi-modal** model that **selectively verifies target signatures** on complex documents, bypassing extraneous parties which increased fraud detection recall by **35%** (maintaining precision) & cut the manual review queue by **50%**
- Engineered a novel Retrieval-Augmented Generation (**RAG**) pipeline with fine-tuned LLMs, which elevated the F1-score for financial document extraction from **0.55 baseline to 0.90 (+0.35)**
- Championed a team-wide shift to **MLOps** best practices, establishing a **CI/CD pipeline** with automated testing raising **unit test coverage to 95%**. This initiative cut **production bugs** by **80%**, decreased deployment failures by **50%**
- Mentored **5 ML engineers** and **30 Business Analysts**, fostering a culture of high-quality software engineering across the team
- Resolved team's most critical Natural Language Processing (NLP) project blockers on GCP base DocAI platform, unblocking **10+ major issues** within 3 months and accelerating project delivery by **25%**

Security Researcher | Bug Bounty Hunter (Web Apps, API) | *Bugcrowd | Hackerone* | Remote **Jun 2021 - Jul 2023**

- Mitigated ~\$1M+ in business risk by identifying 30+ critical system vulnerabilities, preventing potential data breaches affecting over 7M users
- Engineered Python automation tools on Docker/Kubernetes with optimized algorithms, boosting manual testing efficiency by 60% by identifying architectural gaps and potential security biases faster

PROJECTS / RESPONSIBILITY

Smart Traffic Management System | GANs, Reinforcement Learning (RL) and Computer Vision (CV) **Jan 2026 - Present**

- Designing an AI-agent based traffic control prototype integrating YOLO-based video detection and multi-agent RL in a SUMO simulation to optimize urban signal timings
- Targeting **~10% commute time reduction** in average commute time across simulated intersections via real-time data streaming, state aggregation, and RL training pipelines using Python, PyTorch, OpenCV, Cursor-AI and SUMO-RL

Image-Based DocuSearch | Python, CV, Tesseract OCR, NLTK, Scikit-learn, Elasticsearch, FastAPI **Oct 2025 - Dec 2025**

- Built an OCR search engine to **index 100,000+** scanned documents, boosting search relevance by **+0.20 nDCG@10** with a custom NLP pipeline to cleanse noisy OCR data
- Achieved **p95 query latency under 150ms** using an Elasticsearch and BM25 architecture

Teaching Assistant | Mentor - HAWK-i Labs **Aug 2021 - Present**

- Mentored 1000+ students in Computer Science, leading workshops on Python, Machine Learning (ML), Computer Vision and NLP

OPEN SOURCE CONTRIBUTION

GitHub: Maintained 10+ ML repositories and 1 App ([iSee](#))

Research: Reproduced DeepMind's Alphafold research papers from scratch to assist a POC for *Quantiphi Analytics Solutions pvt. ltd.*

EDUCATION

B.E. in Computer Engineering | FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING **Graduated | May 2023**

Relevant Coursework: Machine Learning, Deep Learning, Neural Networks, Computer Vision, Recommendation System, Data Structure & Algorithm

TECHNICAL SKILLS

- Machine Learning:** PyTorch, TensorFlow, Keras, Hugging Face (Transformers), TensorRT, OpenCV, Scikit-Learn
- Artificial Intelligence:** NLP (Prompt Engineering, NLTK), Generative Models (GANs, RAG), Transfer Learning (LoRA), Model Context Protocol, Object Detection (YOLO)
- Data Visualization:** Matplotlib, Seaborn, Plotly, **Data Manipulation & Engineering:** Pandas, NumPy, OpenCV
- ML Optimization:** Distributed Training, Inference, Model compressing, Batch Processing
- Cloud Platforms & Database:** **GCP** (Doc AI, ADK, Vertex AI, GKE, Storage, BigQuery), **AWS** (SageMaker, S3, EC2, RDS)
- LLM Frameworks:** LangChain, crewAI, **Developer Tools & Frameworks:** Docker, Kubernetes, Github, Flask, Django, Postman
- Programming Languages:** Python, Go, SQL (PostgreSQL), Javascript, C++, C, Git, Bash Script, **Ticketing System:** Jira, Rally

CERTIFICATIONS AND TRAINING

Google Cloud Professional Cloud Security Engineer (**GCP PCSE**) | Google Cloud **Dec 2025**

Google Generative AI Leader (**GCP Gen Ai**) | Google Cloud **Nov 2025**

Certified Google Digital Lead (**GCP CDL**) | Google Cloud **Sep 2025**

Certified Google Cloud Professional Cloud Architect (**GCP PCA**) | Google Cloud **Jun 2025**

Certified Google Cloud Professional Machine Learning Engineer (**GCP PMLE**) | Google Cloud **May 2024**

Certified Google Cloud Associate Cloud Engineer (**GCP ACE**) | Google Cloud **Dec 2023**